EE4047 – High Voltage Techniques I

New Topic:

Breakdown in Liquid Dielectrics

Week 11.

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OUTLINE

1. Liquid Dielectrics
2. Breakdown in liquids
3. Factors affecting breakdown in liquids
4. Breakdown types in liquid dielectrics

LIQUID DIELECTRICS

Liquid dielectrics

- Insulation oil
- Transformer oil
- CCl_4 (carbon tetrachloride)

Where to use them?

- Insulation
- Cooling (transfers heat through natural flow in the oil: cold oil replaces heated oil inside a transformer or breaker or other)
- Protection (extinguishing arcs, protecting the metalic surfaces from humidity, dirt and rust)

1 LIQUID DIELECTRICS

Mineral oils are processed products of petroleum. They have Fluorine (F) and Clorine (Cl). These oils may become flamable at high temperatures.

2 BREAKDOWN IN LIQUIDS

- In highly purified liquid dielectrics, breakdown is controlled by phonomena similar to those for gasses and the electric breakdown strength is high (of the order of 1 MV/cm).

- Unfortunately, liquids are easily contaminated and may contain solids, other liquids in suspension and dissolved gasses which are conductive or have less breakdown strength compared to the liquid dielectric they are in.

- Contamination considerably lowers breakdown strength of liquid dielectrics.

- The impact of these impurities is relatively small for short duration pulses (10 s).

- However, if the voltage is applied continuously, the solid impurities line up at right angles to equipotentials and distort the field, so that breakdown occurs at relatively low voltage.

- The line up of particles is a fairly slow process and is unlikely to affect the strength on voltages lasting for less than 1 ms.

2 BREAKDOWN IN LIQUIDS

- Due to the tendency to become contaminated, liquid dielectrics are not usually used alone above 100 kV/cm in continuously energised equipment.

- They are used at much higher stresses (up to 1 MV/cm) in conjunction with solids, which can be made to act as barriers, preventing the line-up of solid impurities and localising of any bubbles which may form.

- The main function of the liquid, when it is used together with solid dielectrics, is to fill up the gaps.

- Breakdown strength depends more on the nature of impurities a liquid dielectric contains than on the nature of the liquid itself.

- These impurities lead to the breakdown of liquid dielectrics below their intrinsic strength.

3 FACTORS AFFECTING BREAKDOWN IN LIQUIDS

1. Temperature (if increases, breakdown voltage **decreases**)
2. Humidity (if increases, breakdown voltage **decreases**)
3. Pressure (if increases, breakdown voltage **increases** too)
4. Solid impurities:dust, dirt, fiber particles (breakdown voltage **decreases**)
5. Gas bubbles (breakdown voltage **decreases**)
6. Electrode geometry, material, surface and polarity (iron has lower breakdown strength than silver)
7. Voltage type, application duration (duration reduces breakdown voltage, DC breakdown voltage is lower than AC)
8. Frequency

FACTORS AFFECTING BREAKDOWN IN LIQUIDS

Impurities

-Impurities which have a breakdown strength lower than the liquid itself (such as bubbles of gas). Breakdown of the impurities may trigger off the total breakdown of the liquid.

-Impurities which are unstable in the electric field (such as globules of water). Instability of the impurity can result in a low resistance bridge across the electrodes, leading to total breakdown.

- Impurities which result in local enhancement of electric field in a liquid (such as conducting particles). The enhanced field may cause local breakdown and initiate complete breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

1. **Electronic breakdown** (based on Townsend Theory in gases, happens in pure liquid dielectrics)

2. **Electromechanical breakdown** (there could be no particles in the liquid, but there may be air bubbles or gas globules, which cause this type of breakdown)

3. **Thermal breakdown** (the particles in the impurified liquids form a bridge and causes breakdown)

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to gaseous inclusions

Gas or vapour bubbles may exist in impure liquid dielectrics, either formed from dissolved gasses, temperature and pressure variations, or other causes.

$$E = \frac{3\varepsilon_1}{2\varepsilon_1 + 1} E_0$$

Field in the gas bubble

Permittivity of the liquid dielectric

Field in the liquid in the absence of gas bubble

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to gaseous inclusions

The electrostatic forces on the bubble cause it to get elongated in the direction of the electric field.

The elongation continues, when sufficient electric field is applied, and at a critical length the gas inside the bubble (which has a relatively lower breakdown strength) breaks down.

This discharge causes decomposition of the liquid molecules and leads to total breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to liquid globule

If an insulating liquid contains in suspension a globule of another liquid, then breakdown can result from instability of the globule in the electric field.

$$E = \frac{3\varepsilon_1}{2\varepsilon_1 + \varepsilon_2} E_0$$

- E: Field in the globule
- ε_1: Permittivity of the liquid dielectric
- ε_2: Permittivity of the spherical globule
- E_0: Field in the liquid in the absence of the globule

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to solid particles

In commercial liquids, solid impurities cannot be avoided and will be present as fibres or as dispersed solid particles.

If the impurity is considered to be a spherical particle of permittivity ε_2 and is present in a liquid dielectric of permittivity ε_1, it will experience a force

$$F = \frac{\varepsilon_2 - \varepsilon_1}{\varepsilon_2 + 2\varepsilon_1} \Delta E^2$$

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due to solid particles

Generally $\varepsilon_2 > \varepsilon_1$, so that the force would move the particle towards the regions of stronger field.

Particles will continue to move in this way and will line up in the direction of the field.

A stable chain of particles would be which at a critical length may cause breakdown.

BREAKDOWN TYPES IN LIQUID DIELECTRICS

The particles can be taken out of the oil by filtering the oil.

Humidity and water bubbles can be eliminated by drying the oil.

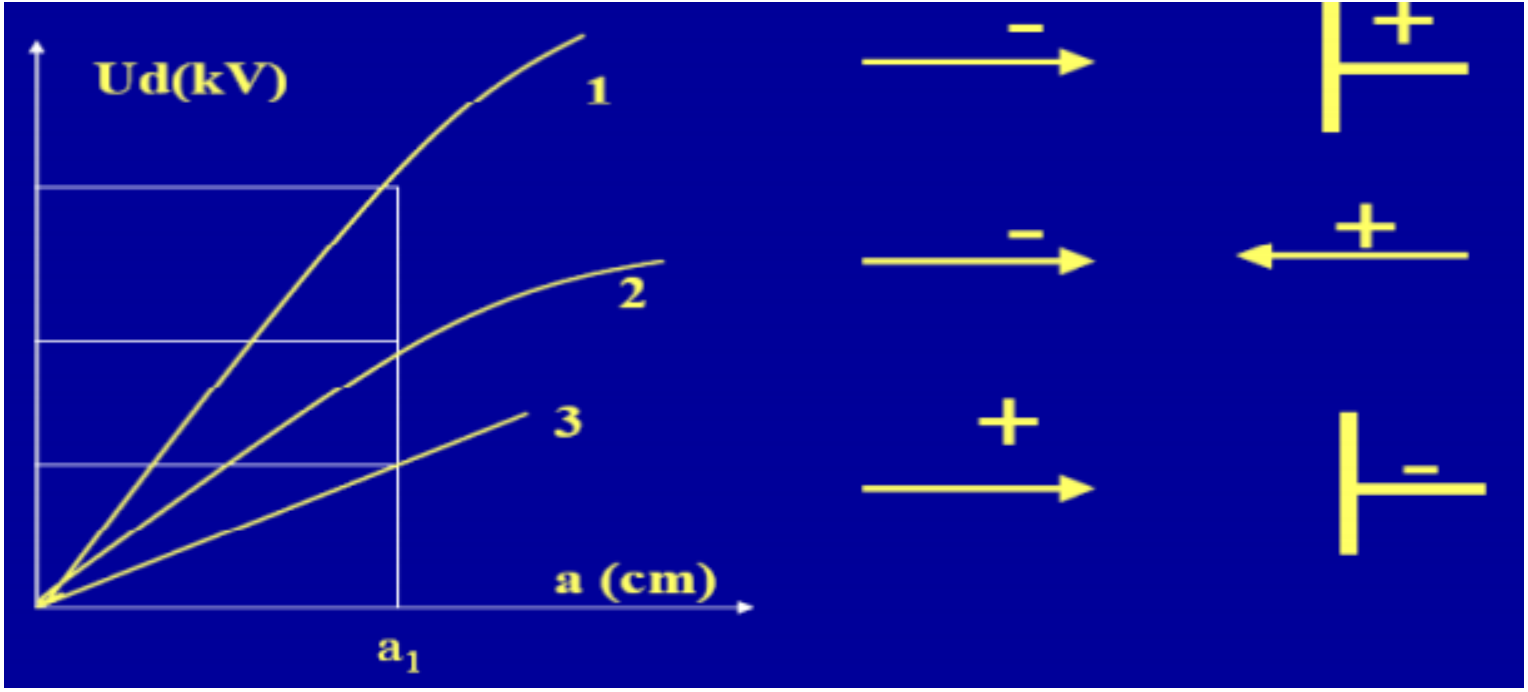
BREAKDOWN TYPES IN LIQUID DIELECTRICS

Desired dielectric oil properties

- Purity (mineral oil with no addition of other materials, also no water-gas-solid particles)
- High flamability point (becomes flamable at a very high temperature)
- High dielectric strength ($E_b > 200\ kV/cm$)
- High liquidity at low temperatures (should not become considerable less liquid due to cold)
- Enough liquidity (low viscosity) at any temperature to extinguish arc
- Should not be very acidic

BREAKDOWN TYPES IN LIQUID DIELECTRICS

Breakdown due electrode polarity and geometry



END OF THE LESSON

Any questions?

Check out the following videos on Youtube:

Breakdown voltage testing of transformer oil: https://www.youtube.com/watch?v=nZAg_A1tb0o

Animation of transformer oil testing: https://www.youtube.com/watch?v=BnrARAP2btk

Transformer oil test on site training: https://www.youtube.com/watch?v=YTp-SjvaA5Q

Oil testing equipment video: https://www.youtube.com/watch?v=WxVKADaLyXk